Project: Worked examples
Title: Pipe pressure drop and pump duty
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1. Scope and method

Calculates the friction and fitting losses of series pipe segments at a given flow, the system curve and the duty point against a pump curve. It also gives the affinity-law shift for a speed or impeller change, and the NPSH.

WHAT IT CALCULATES	RELATION
Design flow and velocity	$Q = \Phi/(\rho\, c_p\, \Delta T)$ when a heat load is given in place of a flow; $v = Q/(\pi D^2/4)$
Reynolds number and friction factor	$Re = \rho v D/\mu$; $f = 64/Re$ for laminar flow ($Re < 2300$); Colebrook–White $1/\sqrt{f} = -2 \log10(\varepsilon/3.7D + 2.51/(Re\sqrt{f}))$ for turbulent flow, by fixed-point iteration (at most 100 steps) from the Swamee–Jain value, which also cross-checks it
Head lost in a segment	$h_f = f\,(L + L_{eq})/D \cdot v^2/2g + K\, v^2/2g$
System curve	$H_{sys}(Q) = \Sigma dz + h_{out} + \Sigma h_{f,lam}\cdot(Q/Q_o) + \Sigma h_{f,other}\cdot(Q/Q_o)^2$, with a laminar segment's friction scaled as Q and every other friction and fitting loss as Q^2 (a closed circuit has $\Sigma dz = 0$)
Water held in the pipes	$V_{pipe} = \Sigma\, \pi/4 \cdot D^2 \cdot L$
Duty point	$H_{pump}(Q) = H_{sys}(Q)$, every crossing found by a scan and refined by bisection, the stable one nearest the design flow reported; pump affinity laws Q ∝ n·D^a, H ∝ $n^2 \cdot D^2$, with a = 1 for a trimmed impeller (the default) and a = 3 for a geometrically similar pump
Net positive suction head available	$NPSH_a = p_{atm}/\rho g + z_s - h_{f,suction} - p_v/\rho g$, with $h_{f,suction}$ that of the 'suction' segments at the larger of the design flow and the duty flow
Shaft torque for the motor	$T = P/\omega$, with P the larger of the design-flow and duty-point shaft power at the speed n_{pump}, or the shifted duty power at the shifted speed $n_{pump} \cdot n_2/n_1$

Basis: Flow of fluids through valves, fittings and pipe[1]; Internal flow: friction factor, minor losses, pump duty[2](Ch. 8, 4th ed.); Flow of fluids in pipes and ducts (principles)[3](§4); Density of liquid water[4].

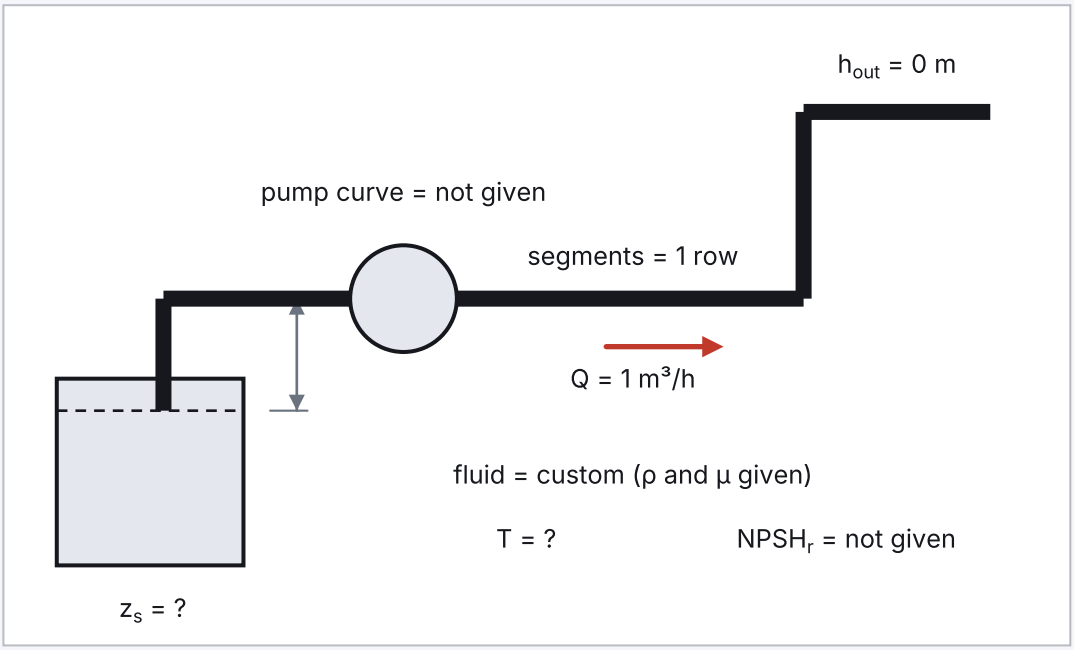


Figure 1 — A pump lifting from a suction level through the pipe segments as listed (three drawn) to an outlet that needs a pressure head. Not to scale; the labels are the inputs as entered.

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
fluid	Fluid	custom (ρ and μ given)		spec: Oil datasheet ISO VG 680 at 20 °C	any
circuit	Circuit (a closed flow-and-return loop has no static lift: the rises entered are ignored)	open (the pump lifts the water by the rises entered)		default	any
segments	Pipe segments in series	1 row(s)		drawing: DRG-L-020 rev A	any
ρ_{custom}	Density (custom fluid)	888	kg/m³	spec: Oil datasheet ISO VG 680	500 … 2000 kg/m³
μ_{custom}	Dynamic viscosity (custom fluid)	800	mPa·s	spec: Oil datasheet ISO VG 680 at 20 °C	0.1 … 1000 mPa·s
h_{out}	Pressure head required at the outlet	0	m	assumption: Open discharge.	0 … 1000 m
Q	Design flow rate (optional; blank = the flow comes from the heat load)	1	m³/h	spec: Transfer pump duty	0.01 … 100000 m³/h
T_{fluid}	Fluid temperature (mean; link the mean water temperature of the sealed system for a heating loop)	20	°C	assumption: Ambient 20 °C.	0 … 100 °C
p_{atm}	Atmospheric pressure (NPSH)	101.325	kPa	default	50 … 110 kPa

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
z_s	Static head at the pump suction (+ flooded, − lift)	0	m	default	-10 … 100 m
v_{limit}	Velocity limit (warning and check)	3	m/s	assumption: Practice limit.	0.5 … 10 m/s
D_{law}	Law for the diameter ratio: a trimmed impeller of the same casing ($Q \propto D$, $H \propto D^2$) or a geometrically similar pump ($Q \propto D^3$, $H \propto D^2$)	impeller trim ($Q \propto n \cdot D$, $H \propto n^2 \cdot D^2$)		default	any
n_{ratio}	Speed ratio n_2/n_1 (affinity)	1		default	0.2 … 2
D_{ratio}	Impeller diameter ratio D_2/D_1 (affinity; the law is chosen above)	1		default	0.7 … 1.1
η_{pump}	Pump efficiency at duty (for shaft power)	0.7		default	0.2 … 0.95

TABLE ROW	VALUES
segments[1]	name = transfer line; L = 40 m; D = 50 mm; eps = 0 mm; K = 0; L_eq = 0 m; dz = 0 m; side = discharge

3. Assumptions

A1. Flow is steady, incompressible and fully developed, and all segments carry the full design flow in series. (Darcy–Weisbach.)
A2. Fitting losses scale with velocity head (K method) or add pipe length (equivalent-length method). (Crane TP-410 practice.)
A3. Both fitting methods may be used on one segment. (Crane TP-410 practice.)
A4. The system curve scales turbulent and transitional friction and all fitting losses with Q^2 from the design flow. ($H_{sys} = H_{static} + k_1 Q + k_2 Q^2$.)
A5. The friction factor is held at its design-flow value on the system curve. ($H_{sys} = H_{static} + k_1 Q + k_2 Q^2$.)
A6. The friction of a laminar segment scales with Q, and the static part is independent of flow. ($H_{sys} = H_{static} + k_1 Q + k_2 Q^2$.)
A7. Water properties are from published correlations at atmospheric pressure, 0 to 100 °C. (Kell 1975; Vogel fit; Hyland & Wexler 1983.)
A8. Any other fluid is entered with its own density and viscosity and their provenance. (Stated.)
A9. h_{out} (Pressure head required at the outlet) is taken as 0 m. (Open discharge.)
A10. T_{fluid} (Fluid temperature (mean; link the mean water temperature of the sealed system for a heating loop)) is taken as 20 °C. (Ambient 20 °C.)
A11. v_{limit} (Velocity limit (warning and check)) is taken as 3 m/s. (Practice limit.)

4. Working

[5] $\rho = \rho_{custom}$
= 888 kg/m³
= 888 kg/m³
Density (as entered)
[6] $\mu = \mu_{custom}$
= 800 mPa·s
= 800 mPa·s

Dynamic viscosity (as entered)

[7] T_{line} = T (no T_{line} entered)

= 20 °C

= 20 degC

Design temperature of the line to insulate: the fluid temperature T, since no T_{line} was entered

[11] Q_{design} = Q

= 1 m³/h

= 1 m³/h

Design flow (as entered)

[12] v_{s1} = transfer line: Q / (π D²/4)

= 1 m³/h / (π × 50² mm² / 4)

= 0.14147 m/s

Velocity, transfer line

[13] Re_{s1} = transfer line: ρ v D / μ

= 888 × 0.14147 × 0.05 / 0.8

= 7.8516

Reynolds number, transfer line

[14] f_{s1} = transfer line: Darcy friction factor

= 64 / Re (laminar)

= 8.1512

Friction factor, transfer line

[15] $h_{f,s1}$ = transfer line: f (L + L_{eq})/D · v²/2g + K v²/2g

= 8.1512 × (40 + 0) / 0.05 × 0.0010204 + 0 × 0.0010204

= 6.6542 m

Head loss, transfer line

[16] $h_{f,total}$ = Σ h_f

= 1 segment(s)

= 6.6542 m

Total friction and fitting loss at the design flow

[18] V_{pipe} = Σ π/4·D²·L

= 1 segment(s)

= 78.54 L

Water held in the pipe segments

[19] H_{static} = Σ dz + h_{out}

= 0 + 0

= 0 m

Static head

[20] H_{sys} = H_{static} + $h_{f,total}$

= 0 m + 6.6542 m

= 6.6542 m

System head at the design flow

[21] dp_{total} = ρ * 9.80665 m/s^2 * $h_{f,total}$

= 888 kg/m³ × 9.80665 m/s^2 × 6.6542 m

= 57.947 kPa

Friction pressure drop at the design flow

[22] dp_{sys} = ρ * 9.80665 m/s^2 * H_{sys}

= 888 kg/m³ × 9.80665 m/s^2 × 6.6542 m

= 57.947 kPa

System pressure at the design flow

[23] P_{hyd} = ρ * 9.80665 m/s^2 * Q_{design} * H_{sys}

= 888 kg/m³ × 9.80665 m/s^2 × 1 m³/h × 6.6542 m

= 0.016096 kW

Hydraulic power at the design point

[26] v_{peak} = highest segment velocity
= 0.14147 m/s
= 0.14147 m/s
Highest velocity
[27] u_v = v_{peak} / v_{limit}
= 0.14147 / 3
= 0.047157
Velocity utilisation

5. Results

SYMBOL	VALUE	DESCRIPTION
Q_{design}	1 m³/h	Design flow
H_{static}	0 m	Static head
V_{pipe}	78.54 L	Water held in the pipes
H_{sys}	6.654 m	System head at Q
$h_{f,total}$	6.654 m	Σ h_f
dp_{total}	57.95 kPa	Friction Δp
dp_{sys}	57.95 kPa	System Δp
P_{hyd}	0.01610 kW	Hydraulic power
v_{peak}	0.1415 m/s	Highest velocity (governing)
T_{line}	20 °C	Line design temperature
v_{s1}	0.1415 m/s	Velocity, transfer line
Re_{s1}	7.8516	Re, transfer line
f_{s1}	8.1512	f, transfer line
$h_{f,s1}$	6.654 m	h_f, transfer line

6. Checks (governing first)

RESULT	CHECK	DEMAND	CAPACITY	UTILISATION	REF
PASS	Highest velocity against limit	0.1415 m/s	3 m/s	0.05	

7. Applicability

WARNING segments: transfer line is laminar (Re = 7.8516), so f = 64/Re. The system curve and the duty point scale its friction with Q, not Q². [laminar]

Defaults used (inputs left empty):
circuit: not entered, so the default of 'open (the pump lifts the water by the rises entered)' is used.
p_{atm}: not entered, so the default of 101.3 kPa is used.
z_s: not entered, so the default of 0 m is used.
D_{law}: not entered, so the default of 'impeller trim (Q ∝ n·D, H ∝ n²·D²)' is used.
n_{ratio}: not entered, so the default of 1 is used.
D_{ratio}: not entered, so the default of 1 is used.
η_{pump}: not entered, so the default of 0.7 is used.

8. Verification

Template bs.pipe-pump@1.10.0 Content hash c67e163da77aac31 Engine calc-engine/1.0.0
Known-answer cases passed 5/5 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0

9. References

1. Crane Co. Flow of Fluids Through Valves, Fittings and Pipe. Technical Paper No. 410. Crane Co.
2. Çengel YA, Cimbala JM. Fluid Mechanics: Fundamentals and Applications. 4th ed. McGraw-Hill Education; 2018.
3. Chartered Institution of Building Services Engineers. CIBSE Guide C: Reference Data. CIBSE; 2007.
4. Kell GS. Density, thermal expansivity, and compressibility of liquid water from 0° to 150°C: correlations and tables for atmospheric pressure and saturation reviewed and expressed on 1968 temperature scale. J Chem Eng Data. 1975;20(1):97-105.